

Prescription Pain-Reliever Misuse Among U.S. Adults

A Data-Driven Analysis Using the 2024 National Survey on Drug Use and Health

Business Objective

Prescription pain-reliever misuse remains an important behavioral-health and public-health concern. Organizations involved in prevention, behavioral-health services, population health, and resource planning may benefit from understanding which characteristics tend to occur more frequently among adults reporting misuse.

The objective of this analysis was to identify demographic, socioeconomic, health, mental-health, and substance-use characteristics associated with reported past-year prescription pain-reliever misuse and evaluate whether those characteristics could be used to distinguish respondents reporting misuse from those who did not.

The goal was not to create a clinical screening tool or predict future individual behavior. Instead, the analysis was designed to identify population-level patterns that could support further research, prevention planning, and more focused investigation of higher-risk behavioral-health characteristics.

Business Question

Which demographic, socioeconomic, health, mental-health, and substance-use characteristics are most strongly associated with past-year prescription pain-reliever misuse among U.S. adults, and how effectively can a classification model distinguish respondents who report misuse from those who do not?

Data Source and Scope

The analysis used the **2024 National Survey on Drug Use and Health (NSDUH) Public-Use File**, published by the Substance Abuse and Mental Health Services Administration.

The original public-use dataset contained **58,633 respondents and more than 2,600 variables**. The analysis was restricted to adults age 18 and older. After data cleaning, the final analytical sample contained **47,274 adults**.

The target variable identified whether a respondent reported past-year prescription pain-reliever misuse. Approximately **3% of the unweighted adult analytical sample** reported misuse, creating a substantially imbalanced classification problem.

Candidate predictors were selected through review of the NSDUH codebook rather than automatically including all available variables. Selected features represented demographic, socioeconomic, health, mental-health, functional, and substance-use characteristics.

Analytical Approach

Data preparation included validating respondent identifiers, evaluating missing values and special survey codes, using official imputation-revised NSDUH variables where appropriate, and creating missingness indicators for selected mental-health measures.

Exploratory analysis was followed by statistical association testing using techniques including chi-square tests, Cramér's V, point-biserial correlation, and Mann-Whitney U tests.

Several classification models were then evaluated, including logistic regression, random forest, and AdaBoost. Because prescription pain-reliever misuse represented only a small percentage of the sample, model performance was evaluated using precision, recall, F1 score, ROC-AUC, PR-AUC, and confusion matrices rather than relying on accuracy alone.

A separate validation partition was used to evaluate class weighting and decision thresholds before fitting the final expanded models.

Key Findings

Substance-use characteristics showed some of the clearest relationships with reported prescription pain-reliever misuse.

Among respondents with selected substance-related conditions, the unweighted misuse rates were:

Substance-Related Condition Past-Year Pain-Reliever Misuse Rate

Alcohol use disorder	7.56%
Marijuana use disorder	8.51%
Nicotine dependence	10.82%
Cocaine use disorder	34.64%
Methamphetamine use disorder	37.03%

Methamphetamine and cocaine use disorder were identified during a targeted second review of the NSDUH codebook. After being incorporated into the expanded model, they became the two largest positive logistic-regression coefficients among the evaluated predictors.

A descriptive substance-risk count based on nicotine dependence, alcohol use disorder, and marijuana use disorder also revealed a stepwise pattern. Reported prescription pain-reliever misuse increased from approximately **1.75% among respondents with none of the three indicators to 25.57% among respondents with all three.**

Mental-health and functional characteristics also showed associations with misuse. Higher psychological distress and functional impairment were associated with higher reported misuse, although their individual relationships were more modest than the strongest substance-use indicators.

Broad demographic characteristics generally provided less discriminating information than substance-use and behavioral-health measures.

Model Results

The final expanded logistic-regression model produced the strongest overall performance among the models evaluated.

Metric	Logistic Regression	Random Forest
Accuracy	0.941	0.929
Precision	0.184	0.144
Recall	0.273	0.263
F1 Score	0.220	0.186
ROC-AUC	0.737	0.731
PR-AUC	0.147	0.117

At the validation-selected classification threshold of 0.50, the logistic-regression model correctly identified **79 of 289 respondents reporting misuse** in the evaluation partition, while producing 351 false-positive classifications.

Although the overall accuracy was high, accuracy is not particularly informative in this setting because approximately 97% of respondents belonged to the non-misuse class. The precision, recall, F1 score, and PR-AUC provide a more useful view of performance.

The results show that meaningful signal exists in the selected predictors, but the model is not sufficiently precise or sensitive to support individual-level clinical decision-making.

Business Interpretation

The clearest finding from the analysis is that **behavioral-health and substance-use characteristics carried more useful information about reported prescription pain-reliever misuse than broad demographic characteristics alone.**

For organizations conducting population-level analysis, this suggests that prevention or outreach research may be more informative when it considers patterns of co-occurring substance-use and psychological-health characteristics rather than relying primarily on age, income, employment status, or other general demographic variables.

The strong descriptive differences observed for methamphetamine use disorder, cocaine use disorder, nicotine dependence, marijuana use disorder, and alcohol use disorder also highlight the importance of examining substance-use patterns collectively rather than treating prescription pain-reliever misuse as an isolated behavior.

These findings should be viewed as signals for additional analysis and resource-planning research rather than criteria for identifying or labeling individual patients.

Limitations

The NSDUH data are cross-sectional, so this analysis cannot determine whether any characteristic causes prescription pain-reliever misuse or predict which respondents will begin misusing medication in the future.

The outcome is rare, which makes positive-class classification difficult and limits both precision and recall.

Survey weights were retained and considered for descriptive population analysis, but the machine-learning models were not fitted as survey-weighted nationally representative prediction models.

Feature selection was targeted rather than exhaustive across all available NSDUH variables. Additional codebook-guided feature exploration could identify useful predictors not included in this version.

The test partition was also examined during iterative model development. Therefore, the reported metrics are best interpreted as development-stage performance rather than results from a completely untouched final holdout.

Finally, this model is an analytical portfolio project and is **not intended for clinical screening, diagnosis, treatment decisions, or individual-level risk assessment.**

Recommendations and Next Steps

A future version of the analysis could expand the codebook review to identify additional health and behavioral-health variables, evaluate repeated or nested stratified cross-validation, reserve a new untouched final holdout, and explore probability calibration.

A dashboard or stakeholder presentation could also translate the statistical results into an interactive population-health view, allowing users to compare misuse patterns across selected behavioral-health and socioeconomic characteristics.

Conclusion

The analysis found that past-year prescription pain-reliever misuse was associated with a combination of behavioral-health, substance-use, functional, and socioeconomic characteristics.

Substance-related conditions—particularly methamphetamine and cocaine use disorder—provided the strongest additional signal in the expanded model.

While the final logistic-regression model demonstrated that these characteristics contain useful classification information, its performance also illustrates the difficulty of identifying a rare behavioral-health outcome from cross-sectional survey data.

Overall, the project demonstrates how large public-health datasets can be transformed from complex survey documentation into interpretable evidence through careful variable selection, statistical analysis, machine-learning modeling, and transparent communication of limitations.